NOISE-RESILIENT QUANTUM-KERNEL CLASSIFIER

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1. Abstract and problem statement

This project evaluates how synthetic gate and readout noise affect a quantum-kernel classifier and tests simple mitigation methods across two connectivity models. Restricted connectivity and imperfect operations can alter kernel estimates and classification results. The experiment compares ideal simulation, noisy simulation, readout mitigation, zero-noise extrapolation (ZNE), and their combination.

2. Dataset and quantum method

The dataset is the guide-specified make_moons(n_samples=100, noise=0.1, random_state=42). A fixed stratified 70/30 split is used. For tractability, the training subset is capped at six examples per class (12 total), while the held-out test partition is retained. Scaling is fitted only on the training partition, and the two features are mapped to angles in [0, pi]. A two-qubit feature map uses single-qubit rotations, CX entanglement, and a data-dependent phase rotation. The compute-uncompute circuit applies U(x) followed by U(z) inverse; the measured P(00) estimates the fidelity kernel K(x,z)=|⟨|^2. A support-vector classifier uses the precomputed training kernel and test-to-training kernel.

3. Processor models and noise

Processor A uses a participant-defined five-node chain. Processor B uses a nine-node graph transcribed from the participant-supplied guide excerpt. Both use basis gates {rz, sx, x, cx}; the synthetic rates are one-qubit depolarizing p=0.001, CX depolarizing p=0.005, and symmetric readout p=0.01. The Processor B graph must be checked against the latest official Challenge Kit before final upload. These are Aer simulation settings, not hardware calibration data.

4. Mitigation and evaluation

Readout mitigation applies symmetric assignment-matrix inversion, followed by clipping and renormalization. ZNE fits a quadratic to kernel probabilities at gate-noise scales 1, 3, and 5 and extrapolates to scale zero while holding readout error fixed. This is noise-parameter scaling, not gate-folding ZNE. The run exports accuracy, precision, recall, F1, runtime, kernel MAE/RMSE, circuit depth, two-qubit gates, SWAP count, and reproducibility metadata.

5. Results and interpretation

Run python -m src.run_experiment with the final 4096-shot setting. Insert interpretations from the generated CSV files only; this report intentionally does not fabricate numerical results. Discuss classification scores separately from kernel error: a smaller kernel error does not guarantee better accuracy. Compare depth, two-qubit gates, and SWAP overhead. Because the feature map uses two qubits and can fit on a connected edge in either graph, zero SWAPs or similar resource counts are plausible and should not be presented as proof of general topology superiority.

6. Limitations and conclusion

This is a small-data, finite-shot local Aer simulation using a reduced training subset and a simplified synthetic noise model. It does not establish physical-hardware performance or quantum advantage. The fixed seed, processor JSON files, requirements, source code, metadata, CSVs, and figures support reproducibility. Final conclusions must be based on the successful final run.

Submission portal: https://qffsrmap2026.com/learning/hackathon